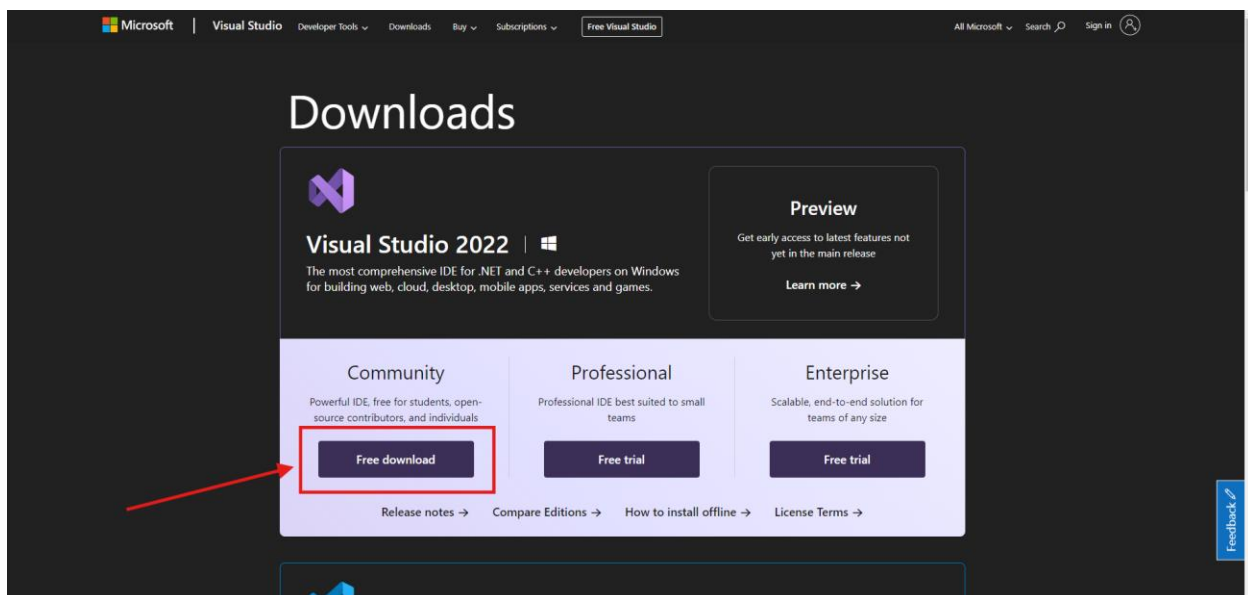


Practice Assignment 1

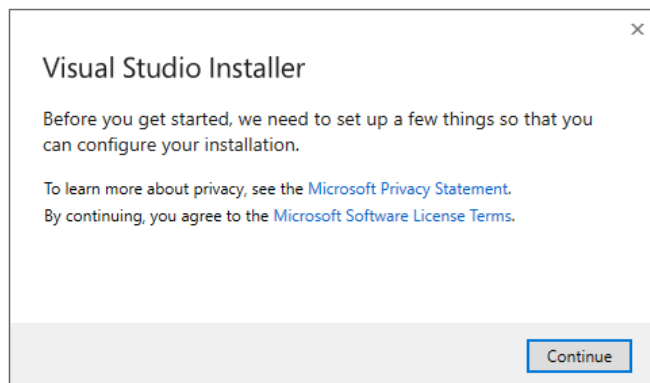
I. Prepare

1.1 Install Visual Studio Community 2022 (Skip if you have already installed it)

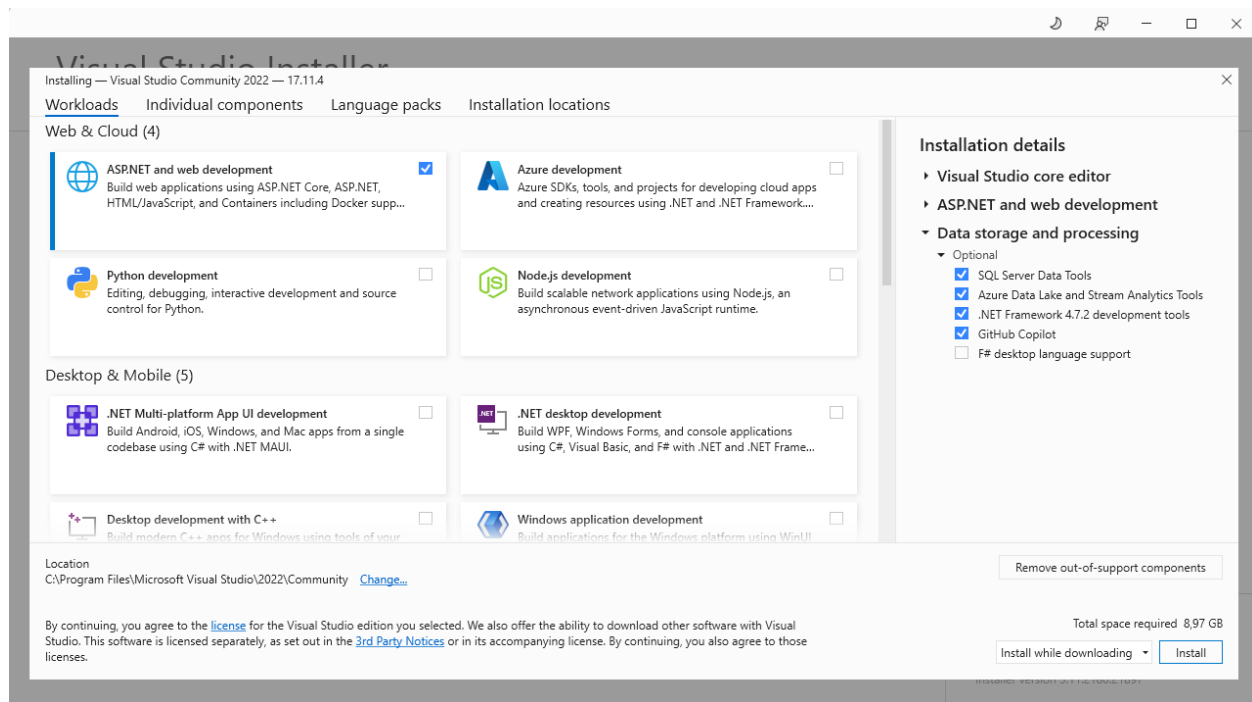
- Go to <https://visualstudio.microsoft.com/downloads/>



- After downloaded, open the file “VisualStudioSetup.exe”  for install.



- Click continue and wait for download the package for install.



- Tick at “ASP.NET and web development”

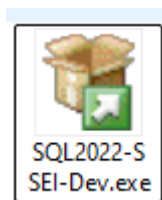
- Tick at “Data storage and processing”

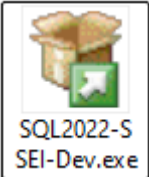
After that click install button and wait until that finish.

1.2 Install the Microsoft SQL Server developer 2022 (Skip if you have already installed it)

- Download the installation file here:

https://drive.google.com/file/d/106Rkzny9Ur5RVCSC0ET-eY4PFjXsVTW/view?usp=drive_link



- After download the file, click to  to install

- The next window will be like this:

Choose the “Basic” version and then install.

- When the installation is complete, click 'Install SSMS' to install the management interface and access the DB engine. Alternatively, you can click the link to download it: Download SSMS: <https://learn.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms?view=sql-server-ver16>

Download SSMS

 [Download SQL Server Management Studio \(SSMS\) 20.2](#) 

Okay! After you finish the preparation, let's do the exercise below.

II. Exercise

Introduction to C# Programming

Objective

- Understand the structure of a C# console program.
- Use variables, operators, conditional statements, loops, and methods.
- Practice writing, compiling, and running a simple C# program in Visual Studio.

Q1. Project Setup

1. Open **Visual Studio** → Create a **Console App (.NET)** project.
2. Name the project:
YourName_StudentID_Lab1
3. The **namespace** must be your name.
4. Create a **public class Program** and a **Main()** method as the entry point.

Q2. Print to Console

In the **Main()** method, use:

```
Console.WriteLine("Hello World");
```

to print text to the screen.

Demonstrates: namespace, class, **Main()** method, and Console usage.

Q3. Add Two Numbers (Methods and Return)

Write a method:

```
static int AddTwoNumbers(int numberA, int numberB)
```

- Return the sum of A and B.
- Call the method from **Main()** and print the result.

Expected Output:

The sum of 5 and 8 is 13

Demonstrates: **methods**, **parameters**, **return values**, and **calling functions**.

Q4. Swap Two Numbers (Variables & Output)

Write a method:

```
static void SwapNumbers(int numberA, int numberB)
```

- Print values before and after swapping.
- Call it inside Main().

Expected Output:

Before swap: A = 10, B = 20

After swap: A = 20, B = 10

Demonstrates: **variables, assignment, and Console output.**

Q5. Student Classification (If–Else)

Write a method:

```
static void ClassifyStudent(double averageScore)
```

Use if, else if, and else:

Score Range Classification

90–100 Excellent

80–89 Good

70–79 Fair

Below 70 Average

Expected Output:

Input: 85

Output: Good

Demonstrates: **conditional logic and comparison operators.**

Q6. Month Information (If + Switch)

Write a method:

```
static void PrintMonthInfo(int month)
```

1. Use if to validate (1–12).
2. Use switch to print month name and number of days.

Example Output:

Input: 5

Output: May - Have 31 days.

Input: 19

Output: The month input is invalid.

Demonstrates: **switch statement** and **flow control**.

Q7. Sum of Numbers (Loop + Validation)

Write a method:

```
static void SumToN(int n)
```

- Check if $n > 0$.
- Use a for loop to sum from $1 \rightarrow n$.
- Print the result.

Example Output:

Input: 5

Output: The sum from 1 to 5 is 15

Demonstrates: **loops**, **input validation**, and **incremental logic**.

Q8. Comment Your Code

- Use **single-line** (`//`) and **multi-line** (`/*...*/`) comments to explain your logic.
Demonstrates: proper code documentation.

Q9. Bonus

Add a simple **user input** feature using:

```
Console.ReadLine()
```

For all question above to make your program interactive.